Table 2: **GSM8K and MedQA Question Answering KV Cache Memory Usage.** HASHEVICT maintains a binary hash matrix of attention keys in memory and, therefore, has slightly higher memory usage than L_2. Our implementation uses 8-bits for binary values instead of 1-bit. Using 1-bit binary numbers will reduce the memory overhead of HASHEVICT by a factor of 8 and decrease the difference in memory usage between HASHEVICT and L_2.

		GSM8K		MedQA	
Cache Budget (%)	Strategy	Compression Ratio	Cache Memory (GB)	Compression Ratio	Cache Memory (GB)
10	L_2	0.8355	0.7603	0.9289	2.5342
	HASHEVICT	0.8380	0.8120	0.8812	2.6338
30	L_2	0.6234	1.7740	0.6957	7.3492
	HASHEVICT	0.6018	1.8531	0.6360	7.5786
50	L_2	0.3968	2.7876	0.4175	12.1641
	HASHEVICT	0.3716	2.8941	0.3901	12.5235
70	L_2	0.1967	3.8013	0.1803	17.2325
	HASHEVICT	0.1857	3.9351	0.1740	17.7285
90	L_2	0.0859	4.8150	0.0498	22.0474
	HASHEVICT	0.0823	4.9761	0.0483	22.6734
100	Full	0.0000	12.6934	0.0000	51.1181

70B-Instruct deployment with 80 layers, 8 KV-heads, sequence length of 8192, batch size of 8 and 50% cache budget, hash dimension of 8-bits, we have that 16-bits and 32-bits only use an extra 20MB, 40MB, and 80MB respectively, which are significantly smaller than the KV cache size of 640GB. Detailed results can be found in Table 10 of Appendix B.

4.8 Attention Loss Ratio

Table 3: **Attention Loss Ratio** measured at 50% cache budget on the GSM8K question answering dataset. HASHEVICT achieves lower attention loss ratio compared to L_2 and even the attention based Scissorhands.

Strategy	Attention Loss
HASHEVICT	0.0336
L_2	0.0340
Scissorhands	0.0448
H_2O	0.0139

To empirically verify that HASHEVICT selects and evicts tokens of low relevance, we measured the average attention loss of the attention heads for HASHEVICT and compared with L_2, Scissorhands and H_2O. Attention loss is defined as the sum of the attention probabilities for the evicted tokens. Or equivalently, 1 - the sum of the attention probabilities for the tokens in the compressed cache. The attention loss ratios were measured at 50% cache budget using prompts from the GSM8K question answering dataset. As per table 3, by accumulating attention score of tokens H_2O has the lowest attention loss ratio as expected. HASHEVICT has slightly lower attention loss compared to L_2 and even the attention-based Scissorhands, proving HASHEVICT's ability of discarding irrelevant tokens.

5 Discussion & Conclusion

In this paper, we introduce HASHEVICT, a novel attention-free eviction strategy for KV cache compression in transformer-based LLMs. By leveraging locality-sensitive hashing (LSH) to approximate cosine similarity, HASHEVICT dynamically determines which tokens to evict from the cache without performing costly attention calculations. Our experiments demonstrate that HASHEVICT can achieve 30-70% compression of the KV cache while maintaining strong performance across various tasks, including free-response Q&A, multiple-choice Q&A, and long-context retrieval.

The key advantage of HASHEVICT lies in its ability to efficiently compress the KV cache pre-attention, enabling significant memory savings and faster inference times. Compared to traditional strategies like L_2 norm-based eviction [Devoto et al., 2024], HASHEVICT excels particularly in reasoning and multiple-choice tasks, where maintaining a diverse set of tokens in the cache is crucial for generating accurate and coherent responses.

There are several potential areas for future work. Investigating hybrid approaches that combine LSH-based eviction with attention-based mechanisms such as [Zhang et al., 2024a, Ge et al., 2023] could offer a middle ground between computational efficiency and retention of high-importance tokens. Further, reducing the overhead associated with maintaining binary hash codes (e.g., by optimizing bit precision) could further enhance the applicability of HASHEVICT to memory-constrained environments.

Acknowledgments

Liu and Fermüller are supported by the NSF under award BCS 2318255 and the National MS Society under award RG-2110-38460. Huang is supported by DARPA Transfer from Imprecise and Abstract Models to Autonomous Technologies (TIAMAT) 80321, National Science Foundation NSF-IIS-2147276 FAI, DOD-AFOSR-Air Force Office of Scientific Research under award number FA9550-23-1-0048, Adobe, Capital One and JP Morgan faculty fellowships.

References

Wayne Xin Zhao, Kun Zhou, Junyi Li, Tianyi Tang, Xiaolei Wang, Yupeng Hou, Yingqian Min, Beichen Zhang, Junjie Zhang, Zican Dong, et al. A survey of large language models. *arXiv preprint arXiv:2303.18223*, 2023.

Jason Wei, Yi Tay, Rishi Bommasani, Colin Raffel, Barret Zoph, Sebastian Borgeaud, Dani Yogatama, Maarten Bosma, Denny Zhou, Donald Metzler, et al. Emergent abilities of large language models. *arXiv preprint arXiv:2206.07682*, 2022.

Dzmitry Bahdanau. Neural machine translation by jointly learning to align and translate. *arXiv preprint arXiv:1409.0473*, 2014.

Minh-Thang Luong. Effective approaches to attention-based neural machine translation. *arXiv preprint arXiv:1508.04025*, 2015.

A Vaswani. Attention is all you need. *Advances in Neural Information Processing Systems*, 2017.

Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Amy Yang, Angela Fan, et al. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.

Yao Fu. Challenges in deploying long-context transformers: A theoretical peak performance analysis. *arXiv preprint arXiv:2405.08944*, 2024.

Suyu Ge, Yunan Zhang, Liyuan Liu, Minjia Zhang, Jiawei Han, and Jianfeng Gao. Model tells you what to discard: Adaptive kv cache compression for llms. *arXiv preprint arXiv:2310.01801*, 2023.

Zhenyu Zhang, Ying Sheng, Tianyi Zhou, Tianlong Chen, Lianmin Zheng, Ruisi Cai, Zhao Song, Yuandong Tian, Christopher Ré, Clark Barrett, et al. H2o: Heavy-hitter oracle for efficient generative inference of large language models. *Advances in Neural Information Processing Systems*, 36, 2024a.

Zhenyu Zhang, Shiwei Liu, Runjin Chen, Bhavya Kailkhura, Beidi Chen, and Atlas Wang. Q-hitter: A better token oracle for efficient llm inference via sparse-quantized kv cache. *Proceedings of Machine Learning and Systems*, 6: 381–394, 2024b.

Alessio Devoto, Yu Zhao, Simone Scardapane, and Pasquale Minervini. A simple and effective l_2 norm-based strategy for kv cache compression. *arXiv preprint arXiv:2406.11430*, 2024.

Zhiyu Guo, Hidetaka Kamigaito, and Taro Watanabe. Attention score is not all you need for token importance indicator in kv cache reduction: Value also matters. *arXiv preprint arXiv:2406.12335*, 2024.

Sifan Long, Zhen Zhao, Jimin Pi, Shengsheng Wang, and Jingdong Wang. Beyond attentive tokens: Incorporating token importance and diversity for efficient vision transformers. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10334–10343, 2023.

Michel X Goemans and David P Williamson. Improved approximation algorithms for maximum cut and satisfiability problems using semidefinite programming. *Journal of the ACM (JACM)*, 42(6):1115–1145, 1995.

Moses S Charikar. Similarity estimation techniques from rounding algorithms. In *Proceedings of the thiry-fourth annual ACM symposium on Theory of computing*, pages 380–388, 2002.

Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, et al. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.

Cheng-Ping Hsieh, Simeng Sun, Samuel Kriman, Shantanu Acharya, Dima Rekesh, Fei Jia, and Boris Ginsburg. Ruler: What's the real context size of your long-context language models? *arXiv preprint arXiv:2404.06654*, 2024.

Yushi Bai, Xin Lv, Jiajie Zhang, Hongchang Lyu, Jiankai Tang, Zhidian Huang, Zhengxiao Du, Xiao Liu, Aohan Zeng, Lei Hou, et al. Longbench: A bilingual, multitask benchmark for long context understanding. *arXiv preprint arXiv:2308.14508*, 2023.

Mary Phuong and Marcus Hutter. Formal algorithms for transformers. *arXiv preprint arXiv:2207.09238*, 2022.

Alexandr Andoni, Piotr Indyk, and Ilya Razenshteyn. Approximate nearest neighbor search in high dimensions. In *Proceedings of the International Congress of Mathematicians: Rio de Janeiro 2018*, pages 3287–3318. World Scientific, 2018.

Zichang Liu, Aditya Desai, Fangshuo Liao, Weitao Wang, Victor Xie, Zhaozhuo Xu, Anastasios Kyrillidis, and Anshumali Shrivastava. Scissorhands: Exploiting the persistence of importance hypothesis for llm kv cache compression at test time. *Advances in Neural Information Processing Systems*, 36, 2024a.

Joshua Ainslie, James Lee-Thorp, Michiel de Jong, Yury Zemlyanskiy, Federico Lebrón, and Sumit Sanghai. Gqa: Training generalized multi-query transformer models from multi-head checkpoints. *arXiv preprint arXiv:2305.13245*, 2023.

Zheng Wang, Boxiao Jin, Zhongzhi Yu, and Minjia Zhang. Model tells you where to merge: Adaptive kv cache merging for llms on long-context tasks. *arXiv preprint arXiv:2407.08454*, 2024.

Akide Liu, Jing Liu, Zizheng Pan, Yefei He, Gholamreza Haffari, and Bohan Zhuang. Minicache: Kv cache compression in depth dimension for large language models. *arXiv preprint arXiv:2405.14366*, 2024b.

Nikita Kitaev, Łukasz Kaiser, and Anselm Levskaya. Reformer: The efficient transformer. *arXiv preprint arXiv:2001.04451*, 2020.

Amir Zandieh, Insu Han, Majid Daliri, and Amin Karbasi. Kdeformer: Accelerating transformers via kernel density estimation. In *International Conference on Machine Learning*, pages 40605–40623. PMLR, 2023.

Insu Han, Rajesh Jayaram, Amin Karbasi, Vahab Mirrokni, David P Woodruff, and Amir Zandieh. Hyperattention: Long-context attention in near-linear time. *arXiv preprint arXiv:2310.05869*, 2023.

Amir Zandieh, Majid Daliri, and Insu Han. Qjl: 1-bit quantized jl transform for kv cache quantization with zero overhead. *arXiv preprint arXiv:2406.03482*, 2024a.

Amir Zandieh, Insu Han, Vahab Mirrokni, and Amin Karbasi. Subgen: Token generation in sublinear time and memory. *arXiv preprint arXiv:2402.06082*, 2024b.

Noam Shazeer. Fast transformer decoding: One write-head is all you need. *arXiv preprint arXiv:1911.02150*, 2019.

Coleman Hooper, Sehoon Kim, Hiva Mohammadzadeh, Michael W Mahoney, Yakun Sophia Shao, Kurt Keutzer, and Amir Gholami. Kvquant: Towards 10 million context length llm inference with kv cache quantization. *arXiv preprint arXiv:2401.18079*, 2024.

Ying Sheng, Lianmin Zheng, Binhang Yuan, Zhuohan Li, Max Ryabinin, Beidi Chen, Percy Liang, Christopher Ré, Ion Stoica, and Ce Zhang. Flexgen: High-throughput generative inference of large language models with a single gpu. In *International Conference on Machine Learning*, pages 31094–31116. PMLR, 2023.

Angelos Katharopoulos, Apoorv Vyas, Nikolaos Pappas, and François Fleuret. Transformers are rnns: Fast autoregressive transformers with linear attention. In *International conference on machine learning*, pages 5156–5165. PMLR, 2020.

Foteini Strati, Sara Mcallister, Amar Phanishayee, Jakub Tarnawski, and Ana Klimovic. D\'ej\avu: Kv-cache streaming for fast, fault-tolerant generative llm serving. *arXiv preprint arXiv:2403.01876*, 2024.

Di Jin, Eileen Pan, Nassim Oufattole, Wei-Hung Weng, Hanyi Fang, and Peter Szolovits. What disease does this patient have? a large-scale open domain question answering dataset from medical exams. *Applied Sciences*, 11(14):6421, 2021.

Guangxuan Xiao, Yuandong Tian, Beidi Chen, Song Han, and Mike Lewis. Efficient streaming language models with attention sinks. *arXiv preprint arXiv:2309.17453*, 2023.

Rewon Child, Scott Gray, Alec Radford, and Ilya Sutskever. Generating long sequences with sparse transformers. *arXiv preprint arXiv:1904.10509*, 2019.